

Python Fundamentals

Lab 4

Description: This lab covers OOP Fundamentals

Deadline: 23 December 2025

Submission Instructions : Rename your file with your name for example (MohamedHamed.pdf, MohamedHamed.py or MohamedHamed.ipynb) and submit it via this form: [Python Task](#)

❖ Class Hospital

Attributes: name, location, doctors (list), patients (list)

Methods:

- 1) add_doctor(doctor) : Adds a doctor to the hospital
- 2) add_patient(patient) : Adds a patient to the hospital
- 3) show_doctors() : Prints all doctors' names and specialties
- 4) show_patients() : Prints all patients' names and illnesses

❖ Class Doctor (inherits from Hospital)

Attributes: name, specialty, patients (list of Patient objects under this doctor)

Methods:

- 1) assign_patient(patient): Adds a patient to the doctor's list
- 2) show_patients() : Shows all patients assigned to this doctor

❖ Class Patient (inherits from Hospital)

Attributes: name, age, illness, doctor (assigned doctor)

Methods:

- 1) `assign_doctor(doctor)` : Assigns a doctor to the patient
- 2) `show_info()` : Displays patient details including assigned doctor

! You can check this link [w3schools](https://www.w3schools.com/python/python_oop.asp) and [OOP - Elzero](#)